

ELECTRONIC MAINTENANCE AND REPAIRS
EEE446 NOTE II

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A. Precautions to Be Taken When Testing Electronic Equipment

When working with or testing any electronic equipment, it's always important to be cautious. Whatever type of equipment you're handling, whether simple or complex, it's important to take the right safety precautions.

Working with electricity comes with huge risks that should never be taken lightly. If you're a hobbyist who loves working with electronic components or an electronics professional at your workplace, safety should always come first.

To avoid personal injury, possible damage to equipment or danger of fire, all work on electronic equipment should be conducted following these safety procedures.

General Safety

Before working on any electronics, consider following these basic safety precautions to help reduce any hazards.

- Remove any electronic equipment you're testing or working on from the power source.
- Never assume the power circuit is off. Test and test again with a voltmeter to confirm.
- Remove fuses and replace them only after the power to the circuit is disconnected.
- Don't connect power to a circuit until you're done working on it and rechecked the work.
- Always ensure that all electronics equipment is properly grounded
- If it's damaged, replace it. For instance, replace cables instead of repairing with insulating tape.
- Always use the right electronics repair and maintenance tools.
- Always return covers after removing them to reduce the risk of electric shock.
- Make sure your circuit is not overloaded.
- Always have safety equipment like a fire extinguisher, a basic first aid kit and a mobile phone nearby.

Personal Safety

It's important to ensure that you're safe when working on electronic circuits. Here are some personal safety precautions to keep in mind:

- Always keep your work area dry.
- Always work in a well-ventilated area.
- Don't wear flapping or loose clothing when working.
- Don't work with metallic jewelry on your hands like watches, rings and bracelets.
- Don't use bare hands to remove hot parts.
- Always wear non-conductive shoes.
- Always wear insulator gloves in your hands when carrying out repairs.
- When removing high-voltage charges on capacitors, always use a shorting stick.
- Don't hold the test prods when measuring voltage over 300V.
- Always remove power to a circuit before connecting alligator clips.
- Always wear safety goggles.
- Be careful when handling large capacitors as they can still hold high voltage even after you've disconnected the circuit from power.

High Voltage Safety

One mistake that electronics experts make when doing repairs or maintenance work is assuming routine safety procedures after getting all too familiar with their work. It's important to know

that most electronic equipment use high-voltage that is dangerous and can be fatal. Always follow these safety precautions when working on or near high-voltage circuits.

- Don't work on electronic equipment or make repairs with high voltage on.
- Don't take chances doing what you're not sure about.
- Consider using an isolation transformer when working on AC powered electronic circuits or equipment.
- Never tamper with interlocks.
- Don't ground yourself: Make it a practice to use only one hand when connecting equipment to an electronic circuit.

Fire Safety Precautions

When working with electronic equipment, there is often a risk of fire caused by a short circuit or other reason. Follow these precautionary steps:

- Avoid anything that would cause a fire around your working area like paper, cloth or other combustible materials.
- Look out for damaged wire insulation, overheating of electronic equipment, damaged circuit boards and corrosive components like batteries.
- If there is a burning smell on your electronic equipment, disconnect the power source.
- If there is a fire, use a nonconducting dry powder or CO2 fire extinguisher.
- Always check your circuit to be sure that everything is okay after repairs or maintenance before connecting power.

Electric Shock

One of the major hazards when working with electronic equipment is electric shock. To avoid this, you should take a few safety precautions, including:

- Always read safety procedures that come with every electronic equipment you're about to test or work on.
- Recheck all wires for bad connections
- Always make sure that all parts of electronic equipment are well-mounted to prevent accidents.
- Keep electronic equipment away from water and other liquids
- Always check for signs of wear, defects and fraying on electronic equipment cables, cords and connectors.
- Use special safety rubber gloves and shoes.

Testing Equipment

With the increasing use of electronics in homes and workplaces, safety is becoming more and more important to consumers and service experts. If you're an electronics expert offering repair and maintenance services, it's important to invest in the right new or used testing equipment for your work—that ensures that you're able to carry out your job safely.

B.1 Alignment

Alignment is the process of taking an equipment back to its default or optimal operational conditions or the proper adjustment of the components of an electronic circuit, machine, etc., for coordinated functioning.

B.2 Types and Examples of Alignment

1. Manual alignment: Type that needs human effort like tuning of variable resistors, variable capacitors, adjusting of values in microcontroller programs, to correct color balance in Tv receivers, Video and audio speed in Tape recording systems, siren sounds pitch, camera field of view, etc.
2. Automatic alignment: Type that is done using close loop control system like automatic focusing in cameras, video tracking in VCR, automatic frequency control in radio receivers, noise reduction in audio work etc.

B.3 Antenna Alignment

Antennas are normally directive, receiving more signal, or transmitting more signal in one direction than another.

To ensure that the best performance is obtained, it is necessary to align or point the antenna in the right direction. This antenna alignment or orientation is an key part of any antenna installation. Incorrectly aligned and pointing in the wrong direction, the antenna will perform poorly.

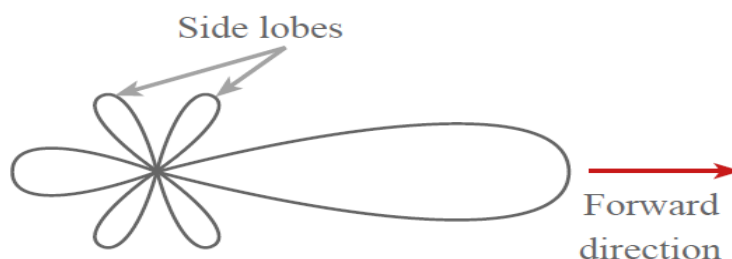
Antenna directivity

Antennas, particularly those used for television reception have a directional pattern and when they are installed, they need to be orientated or pointed towards the television transmitter if the best reception is to be obtained.

Antennas have a directional pattern or polar diagram. For most television antennas they have a marked increase in performance in one direction, and reduced performance in others.

The gain in the main direction enables a much better television signal to be received, but in others it is much reduced. This enables any interference to be reduced in level at the input to the television.

However, this directional pattern means that the antenna must be correctly aligned or orientated, otherwise the main direction of sensitivity will not be aimed at the transmitter and the performance will be reduced.



Accurate alignment of the TV antenna is essential if the best performance is to be obtained. Typically, the longer the antenna, the higher the gain and also the narrower the directional pattern. This makes accurate alignment crucial.

B.4 Methods of aligning an antenna

It may sound as if TV antenna alignment would be an easy job. As the television receiver is normally located well away from the antenna it is not easy to move the antenna, and then check the performance. Small adjustments can take a long time if only one person is present, moving between the antenna and the receiver. Small adjustments can therefore be difficult to make. Even when two people are available, it is still surprisingly difficult.

There are several techniques that can be used to align the antenna:

- *Visual alignment:* This is probably one of the simplest methods of aligning the antenna. When up with the TV antenna, simply point it in the direction of the television transmitter mast. Obviously, this required the TV transmitter mast to be visible, but if not, it is often possible to align the antenna in the same direction as other local antennas. This may be satisfactory in some instances, but signal strengths can vary from one house to the next and with reflections often present, the direction required in one location may be different to that needed in another.
- *Use smartphone app:* There are several smartphone apps which help orientate the antenna towards the television transmitter. There is a good selection, both for Android and also for Apple iOS. Typically, they have a list of the television transmitting antennas with their locations and then they use position location and the orientation capabilities of the smartphone to be able to provide the direction or orientation for the various transmitters. Typically, a list of close stations is provided, as the nearest one may not always be the best.

This method may not always work as well as expected if the television transmitter mast is not directly visible because if the transmitter cannot be directly seen the signal used may be a reflection. It could bounce off buildings, hills and the like. In this way it is best to ensure that all options are taken to get the best signal and picture quality.

- *Signal strength meter:* There are several low-cost TV antenna signal strength meters or alignment meters that can be used. These antenna alignment signal strength meters are straightforward and can enable the easy alignment of the antenna actually at the point of the antenna if it is possible to access the cable to connect to the meter.

These TV antenna alignment meters are generally very basic and measure the strength of signal across the whole TV band - they are just a simple analogue RF signal strength meter and typically they give a basic strength indication on a set of LEDs. This enables the antenna to be orientated to obtain the strongest signal.

The TV antenna alignment meters provide a basic indication of the strongest signal, but as they are wide band, they could pick up other signals and provide a false reading.

However, with a bit of intuition they can work reasonably well, although the sensitivity is limited.

When thinking of investing in a TV antenna alignment signal strength meter, remember to check whether it is for satellite television antenna alignment, or terrestrial television antenna alignment. As signals from these two different types of transmission are on vastly different frequencies, different alignment meters are needed. Some TV antenna alignment meters are able to accommodate both terrestrial and satellite and are advertised as such.

Direction of maximum gain

It is obviously important to know which way any antenna will have its direction of maximum gain. Most terrestrial antennas are based on the Yagi array.

This has a reflector at the back and a series of directors at the front. The element to which the feeder is connected is often what is termed a folded dipole and somewhat larger than the others. The reflector is right at the back, followed by the driven element and a series of directors progressively getting slightly smaller.



Yagi Uda antenna showing direction of maximum radiation

With a little intuition it is possible to determine the direction in which the antenna should point. This can then be aligned with the direction of the television transmitter.

Bonus tips for TV antenna alignment

Apart from the basic techniques for TV antenna alignment, a few additional tips may help make sure that the best performance is achieved.

- *Don't rely on TV internal meter:* Some televisions have internal antenna alignment meters to help measure signal strength and provide a tool for the TV antenna alignment.

Although this can give an indication of the strength, the way it is done can be a little misleading in some instances. If difficulties are experienced, then it can be easier to rely on an external antenna alignment meter for more accurate antenna alignment, especially if it can be used at the antenna itself. This can mean that a direct indication can be seen at the antenna, and the results of any changes made at the antenna can be seen directly.

- *Use good quality TV antenna cable:* TV antenna cables vary widely in quality. There will always be some loss in the antenna cable, but this should be minimized, especially in areas where the signal level is low because the transmitter is a distance away, or because of shielding by buildings, hills, etc. Cheaper coax will almost always give a higher loss, and it is likely to be more prone to picking up interference. Even small fly leads can introduce loss. Here, cheap fly-leads bought in supermarkets may not have the quality needed. Some have been noted to have noticeable levels of loss, even for very short lengths. Always buy the best you can, and then the performance will be assured.
- *Placement as well as alignment is important:* The placement of an antenna is often just as important, and sometimes more important than accurate alignment. Local terrain, trees, buildings, interference can make the antenna location very important. If the antenna is located externally, then there may not be too much choice in the location, but also the performance is likely to be better, but be aware of local obstructions - do trees, tall poles, buildings, etc. mask a direct line to the transmitter, and would a small move improve the 'radio visibility.'

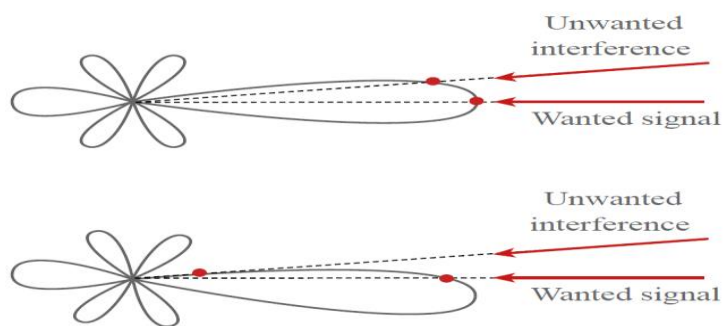
Antennas located internally are more prone to local obstructions, etc., but also there is often more choice in the location. Make sure that the direct line to the TV transmitter is not obstructed by objects like water tanks, solar panels on the roof, etc. Also, the direct line to the TV transmitter may not always be the best direct. If in doubt use a signal meter to check for the best antenna alignment and performance.

- *Check polarization:* The polarization of a television signal is important when aligning an antenna. Get the polarization 90° out and there will be no signal. The polarization of the antenna matches that of the signal it is receiving and is broadly that of the elements of the antenna.



Vertically polarized TV antenna - note the elements are all vertical

- Although most television signals are horizontally polarized and the elements of the antenna are the same, some are vertically polarized. Very occasionally some broadcast stations may be what is termed slant or diagonally polarized, e.g., 45° to the vertical so that both horizontal antennas (often used for home reception) and vertical antennas (in cars and on portable radios) can be used. Check the polarization - this can be done by referring to the local information, or just looking at what all the other local antennas have - and align the antenna polarization accordingly.
- *Keep away from the antenna:* When adjusting the antenna, try not to get too near to it when any adjustments are being made. The close proximity of people to the antenna can change its performance. Absorption, reflections, proximity capacitance effects that change its resonance, and a host of other things can affect the antenna. Definitely do not get in front of the antenna, probably behind it is best, but check that any movements you make do not change the signal meter reading unduly, and then finally tighten the fixings and then check the signal quality when you are well away from the antenna.
- *Adjust for best signal:* Although there is a major move to digital transmission of television, there are still analogue transmitters on the air. Often interference from reflections or other sources can cause patterning on these analogue signals - sometimes called a herring-bone effect. The antenna can be aligned to minimize the interfering signal, whilst retaining an acceptable level of the wanted signal. Another effect seen on analogue signals is a ghost picture where the reflected signal is delayed by a small amount causing second ghost signal to appear.



Aligning TV antenna for minimum interference

It can be seen that aligning the antenna for the highest wanted signal can also enable high levels of an unwanted interfering signal like a reflection. Offsetting the antenna slightly can considerably reduce the unwanted signal whilst only having a minor impact on the wanted one. Very occasionally it may also be useful to slant the polarization of the antenna slightly to reduce the level of the interfering signal.

C.1 Alignment of a Satellite dish

It is important to know that for satellite dish alignment you must have a clear un-obstructed clear view of the satellites in space. This means there can be no trees or buildings in the way or temporary structures like scaffolds and the satellite dish cannot work when installed inside the building, like discreetly in a loft. It would be nice, but not possible. To get a satellite dish aligned correctly, you must maximize the signal on three planes. This is the left and right or east and west (azimuth) adjustment. The elevation angle which is up and down and the skew adjustment, this is an angle that the LNB itself is fixed into the LNB holder.

C.3 Azimuth (East & West) adjustment

Your satellite dish needs to align to the correct position for the satellite TV services you wish to receive and this is represented in degrees off due south, not magnetic south like compasses will give you.

The azimuth is the most important alignment of them all as once you have got this fixed, it is just a case of setting the elevation before you start to find the satellite. If you have the azimuth adjustment off, you could waste hours finding the wrong satellites (there are loads) so you will need to do some investigation on the satellite angles that you wish to use.

C.4 Elevation Adjustment

Once you have the azimuth set correctly so will need to correctly set the elevation angle before you find the satellite, this is again given in an angle, but the absolute correct angle will depend on where you are on the earth. As the telecommunication satellites orbit around the equator, the Astra 2 satellites occupy a geo-stationary orbit position above the Democratic Republic of Congo for example the closer the equator you're the higher in the Sky the satellites will be in the horizon. Until you get to the equator where the satellites will literally be straight up above you, the other extreme of this will be in places like Iceland where the satellites will appear very low on the horizon which can make line of sight much trickier due to nearby obstructions like trees and nearby buildings.

once you have the azimuth set it is far easier getting the elevation right rather than the other way around. Most satellite dishes usually have on them an elevation angle setting which will help you get the right sort of region for the elevation angle setting, this angle on the dish itself assumes that the dish has been mounted on a perfectly flat, vertical surface which is quite uncommon. Many dishes also have a spirit level in built.

C.5 Skew Adjustment/ LNB setting

The final alignment that needs to be set is the skew adjustment on the LNB. This is the angle that the LNB itself sits within the LNB holder. Unless the satellites that you wish to align your dish to are perfectly due south (or north down under!) then you are going to need to make an adjustment for the LNB skew. The reason this is so important is because satellite signals are broadcast in both horizontal and vertical polarizations and without setting the skew correctly the LNB and satellite dish will not be able to optimally identify the difference between these and they can interfere with one and another resulting in an unreliable signal, picture pixilation and possible complete loss of signal altogether.

C.6 Tools to Help You Find the Satellites

1.Kingofsat – Information on Satellite TV Services

On it you can get all the information on TV services that you will ever need on satellite TV services. Use it for information on satellite angles, frequencies, packages, encryptions and much more

2.What Satellite Magazine (Wotsat) – Information on Dish Sizes

Here in the back of every of every edition is a complete satellite list and dishes sizes that are required. You may not know this already, but different satellites will require different satellite dish sizes for different satellites. The weaker the satellite signal is in the location where you wish to install the dish, the larger dish you will require for reception from that satellite as larger satellite dishes pick up more signal.

3.Dishpointer App – Help Identify the Satellite Arc

There are lots of satellite alignment apps out there, free and paid. I believe that the Dish pointer App costs around £10 but what this will allow you to do, when calibrated correctly, is to hold up your phone towards the satellites and using the camera and your geographic co-ordinates it will map out on screen where the satellites appear in the Sky on the horizon. This can help you visualize where the satellites are and identify any line-of-sight issues like clearance of a nearby obstacle and is imperative to get right when doing motorized satellite dish installations.

C.6 Compass/ Inclometer

You can beat a good old compass and inclometer and I much prefer using one of these to quickly find a satellite angle. Some of the benefits of using one of these is that you can quickly pull one out and it doesn't require power or a WIFI or mobile internet signal. The one downside to using a compass to find your azimuth angle (east/ west) is that it will direct you to magnetic south and not due south, often these are the same but because the magnetic field around the earth changes there can be as much as few degrees difference which can be catastrophic when aligning satellite dishes

C.7 Satellite Dish Alignment Tool – TV Spectrum Analyzer

This will allow one to see in almost real time the signals and transponders as I move the satellite dish to quickly find the one, I want.

C.7 Satellite Receiver Beeper

Most satellite receivers will give you some sort of signal strength readings, usually this will include a signal reliability or signal quality reading also like on standard Sky boxes and Freesat boxes. Although the bars or percentage readings don't really mean anything, I mean this by the fact that there is no such thing a 100% satellite signal reading that many give you. Signals just aren't read that way. It can be used as a rough guide to show if you have a good or bad signal.

C.8 Tips for Aligning Your Satellite Dish

1. Use Neighbor's Satellite Dish as A Reference

One thing could save you a fair amount of time, certainly picking the correct position would be to use a neighbor's satellite dish as a reference point to where you want to point your own dish. You can't rely on these entirely however as depending which way you're looking two satellites dishes pointing the same way can still look they are off somewhat. Also, you would be assuming that the neighbor's dish is pointing the correct way, it might not be, they might not even use the thing or it could be aligned to satellites that you don't want like those used for foreign TV.

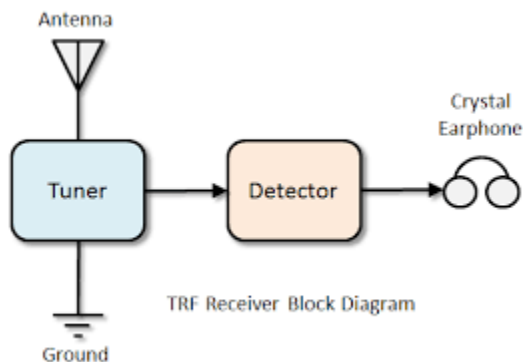
2. Avoid Trees

Although with the natural elevation setting the signals could be beamed comfortably over the top of nearby trees I would say try and avoid them altogether in the satellite line of sight. This may mean that you must install your satellite dish in a different position than you had intended, perhaps with a longer cable run but it could save you lots of bother having to move it at a later date. Trees can cause problems for several reasons:

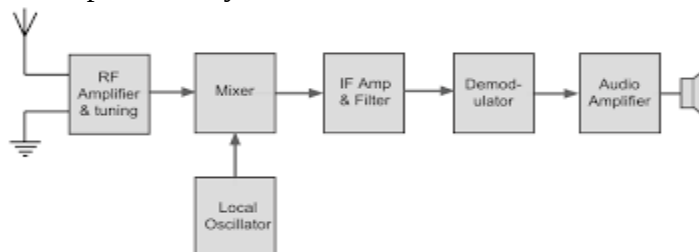
- i- They move – Branches could bow in front of the satellite signal
- ii- They grow – Even if you have a signal today, you might not in a summer or two when the tree has grown
- iii- They leaf in summer – If you are installing a satellite dish pointing through a bald tree, the signal may pass through this fine, but come the summer when the leaves are out, this will most likely be a different story

D.1 Block Diagrams

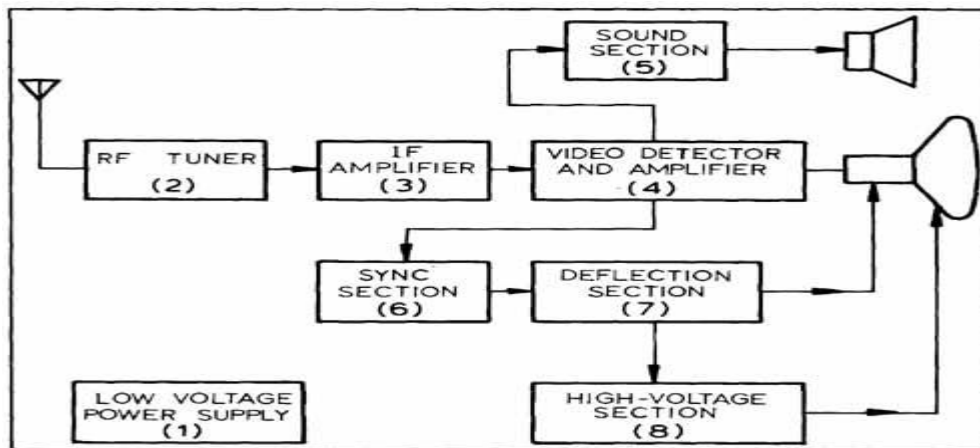
1. Simple radio



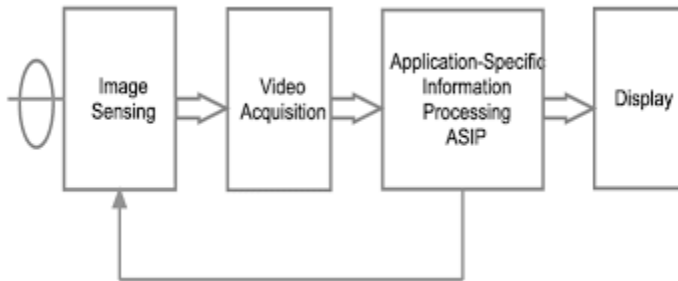
2. Superheterodyne receiver



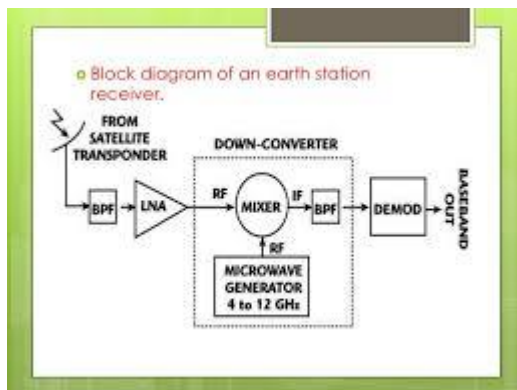
3. TV receiver



4. Video camera



5. Satellite receiver



TUTORIAL QUESTIONS

1. What is the difference between alignment and calibration of electron equipment?
2. Draft the block diagrams of the following i) Heterodyne radio receiver ii) TV receiver iii) Satellite earth station iv) Video camera.
3. Why is it necessary to avoid trees when aligning satellite dishes?
4. Briefly discuss the tools used in satellite dish alignment.
5. Describe the three adjustments that must be undertaken for correct alignment of a satellite dish.
6. What is alignment.
7. State the two types of alignment and give three examples in each case.
8. Briefly describe the three methods for aligning a Tv antenna.
9. How can interference signal be reduced in the TV antenna?
10. List the basic safety precautions to taken when servicing electronic equipment.
11. List the personal safety precautions to be taken when working on electronic equipment.
12. Given a voltmeter and an ammeter only show how the power consumed by a lamp connected across mains be determined.
13. Given a multimeter describe how to determine a transistors terminals and type.
14. What is the function of a signal generator?
15. How can a signal generator be used to troubleshoot a faulty audio circuit?
16. Write short note on diode test.
17. Explain in brief the testing methods necessary to locate the following faults in electronic circuits
 - (i) short circuit
 - (ii) Open circuit
18. Describe briefly how to use a digital multimeter to identify the type/pin configurations of a BJT transistor.
19. Describe in brief how to use a signal generator and oscilloscope only to troubleshoot the two-stage transistor amplifier shown in figure Q.2a.

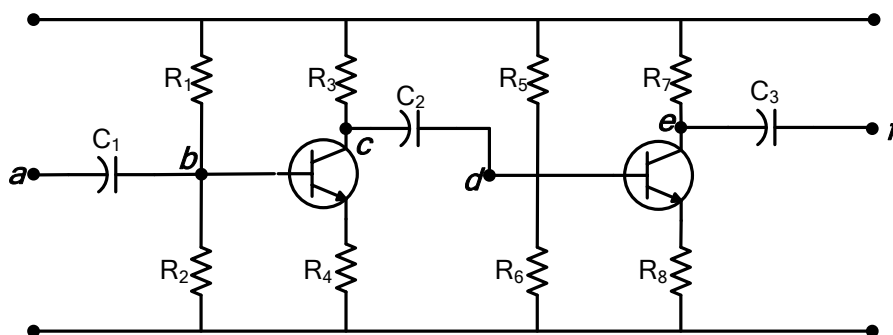
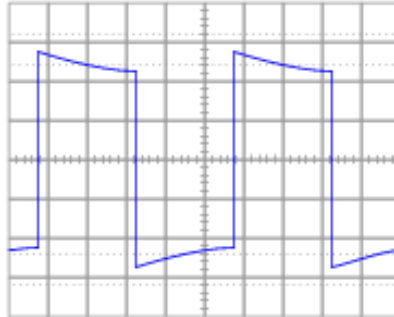


FIGURE Q-2a

In Fig.Q2a above, a pure sine wave of amplitude 1.0 mV is injected at point **A**. Use a diagram to show how the systems output waveform can be observed, also comment on the systems waveform quality and amplitude at points **a**, **b**, **c**, **d**, **e**, and **f** when, system is okay

20. State the safety precautions to be taken during testing of electronic equipment.

21. Fig.Q.3a shows the waveform displayed by an oscilloscope with a vertical setting of 4.0 V per division and a time base setting of 0.75 millisecond per division.



Determine the following:

- (i) frequency (f)
- (ii) V_{peak}
- (iii) $V_{\text{p-p}}$
- (iv) V_{rms}

22. Describe in brief the alignment method for a typical CD player

23. Draw a well labeled 15.0 V dc regulated power supply that use the following components –

- i. Switch
- ii. Fuse
- iii. transformer
- iv. rectifying diodes
- v. Smoothing capacitor
- vi. 15 V regulator

If no output from the power supply, describe briefly the steps you would take to detect the fault.